

Solution Requirements

Requirements Specification - Rising Waters

1. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent (minimum); Intel i5 or above (recommended for model training).
- RAM: 4 GB (minimum); 8 GB (recommended for running local notebooks and training pipelines).
- Storage: 2 GB free disk space (to house datasets, virtual environments, and source code).
- Network: Active internet connection for Vercel/GitHub deployment and initial library installations.

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Development Environment: VS Code or PyCharm.
- Python Runtime: Python 3.8 to 3.12.
- Web Framework: Flask (WSGI web micro-framework).
- Web Libraries: Bootstrap 5, Bootstrap Icons, FontAwesome.
- ML Libraries (Local Development): Scikit-learn, XGBoost, Pandas, Numpy, Openpyxl.

2. Data Flow Diagrams (DFD)

Level 0 DFD: Context Diagram

graph TD

```
User([Meteorologist / Coordinator]) -->|Input Weather Parameters| System[[Rising Waters]]
System -->|Flood Warning / Safe Status| User
System -->|Confidence Score %| User
```

Level 1 DFD: System Architecture

graph TD

```
User([User]) -->|POST Web Form| App[Flask Server app.py]
App -->|Input Sanitization| Val{Validation Engine}
Val -->|If Errors| Flash[Flashed Error Message]
Flash -->|Redirect| User
Val -->|If Valid| Scale[Manual Scaling scaling_loop]
Scale -->|Z-Score Features| Model{Decision Tree Classifier}
Model -->|Binary Class| Predict[Prediction Result]
Model -->|Sigmoid Confidence| Prob[Confidence Score %]
Predict & Prob -->|Render Outcome| Result[chance / no_chance Card]
Result -->|HTML Display| User
```